Developing a Digital System to Monitor Head Clamp Forces During Neurosurgery

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Introduction

The **Head Clamp** (MHC) is a medical device that **holds the head rigid** during neurosurgery, utilizing **3 metal pins** that apply approximately 60-80 lb of force to the skull. An **analog torque screw** is currently used to **communicate the force** exerted on the skull by the prongs.

The **current method** of communicating pressure is **imprecise, difficult to read and often distrused by surgeons** [1],[2]. Additionally, the system provides no information about whether force is evenly distributed across the pins [2]. This can lead to **complications** during surgery, including lacerations, **skull fractures**, and, in severe cases, **brain injury** or death [3].

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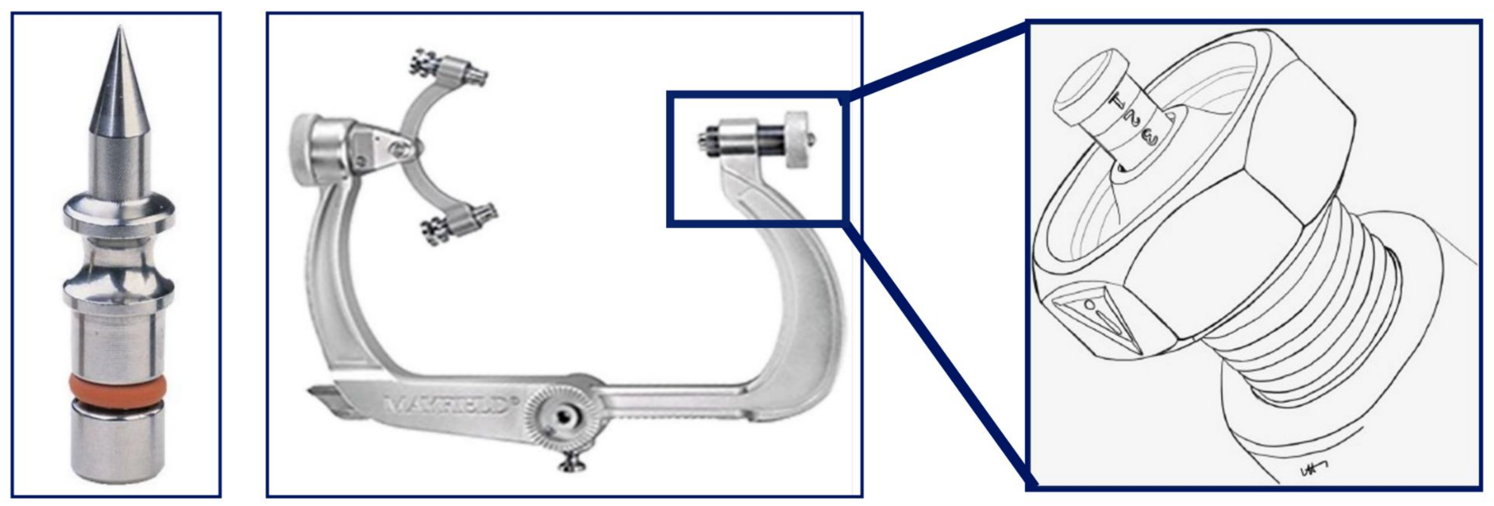

Figure 1. Head clamp pin (left), head clamp (center), analog torque screw (right)

Objective

- The device provides **accurate and intuitive real-time** measurements for the overall force exerted by the clamp and each individual pin.
- The novelty of the device lies in its ability to sense force distributions by capturing **real-time, per-pin force data**.
- Combined with a **digital interface** and **actionable alert system**, our system seeks to increase patient safety by providing **clear, timely feedback** to guide intervention before critical failures occur.

Materials and Methods

Fixture:
A modular sensing fixture was developed to measure per-pin forces without modifying the head clamp. A washer-style load cell was positioned between the rocker arm and skull pin to enable direct axial load measurement. The assembly is stabilized by a steel backing washer and enclosed within a custom SLA-printed housing that prevents lateral motion during tightening.

Electronics
An ESP32 microcontroller was used for data acquisition via its onboard 12-bit ADC. The system was powered by an external DC supply, and firmware written in C++ sampled voltage signals and converted them to force using a calibrated transfer function. Data were transmitted to the user interface via wired serial or Bluetooth Low Energy. No filtering, averaging, or outlier rejection was applied during testing.

UI/Reporting:
Force data are visualized in a real-time interface built with TypeScript, React, and Vite. The display includes total clamp force, individual pin forces, system messages, and an event history panel. User-defined thresholds detect out-of-range forces and pin imbalances, which are highlighted through visual alerts. Time-stamped data are recorded before and after threshold crossings to support retrospective analysis.

System

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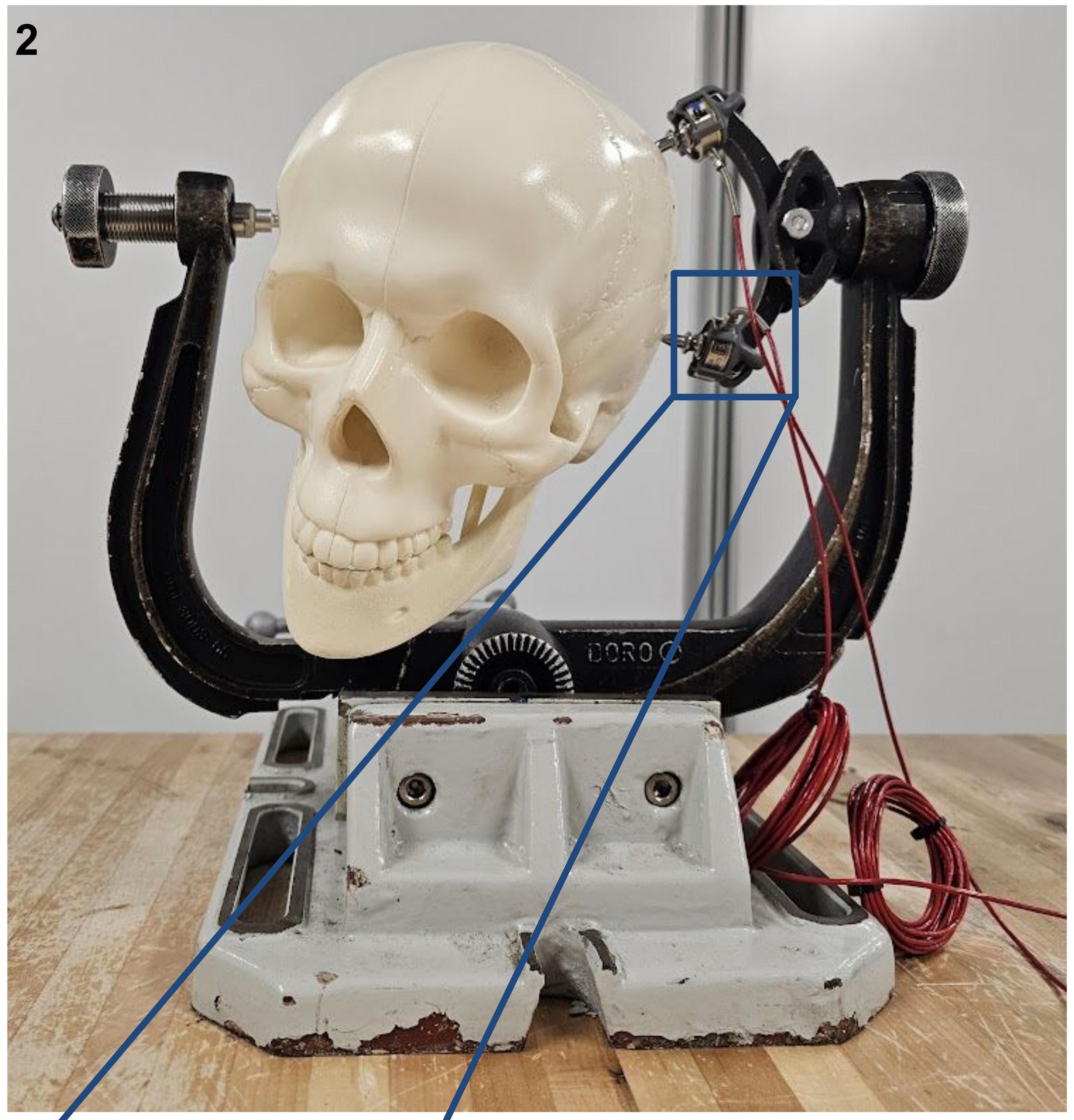

Figure 2. Head clamp assembly showing the device integrated.

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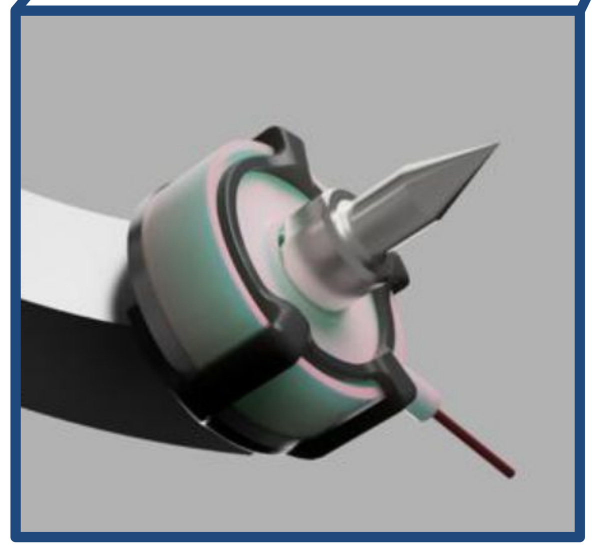

Figure 3. Integration of the device with the head clamp system, illustrating how the pin, clamp, and device interface.

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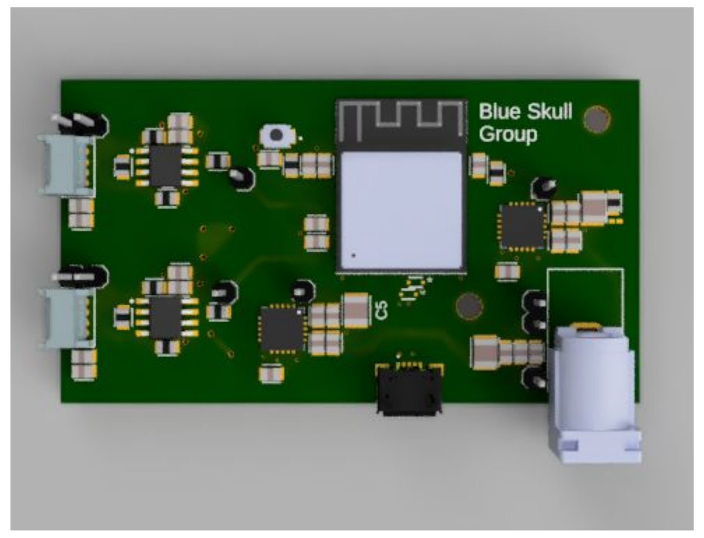

Figure 4. PCB with microcontroller for signal conditioning, processing, and data transmission.

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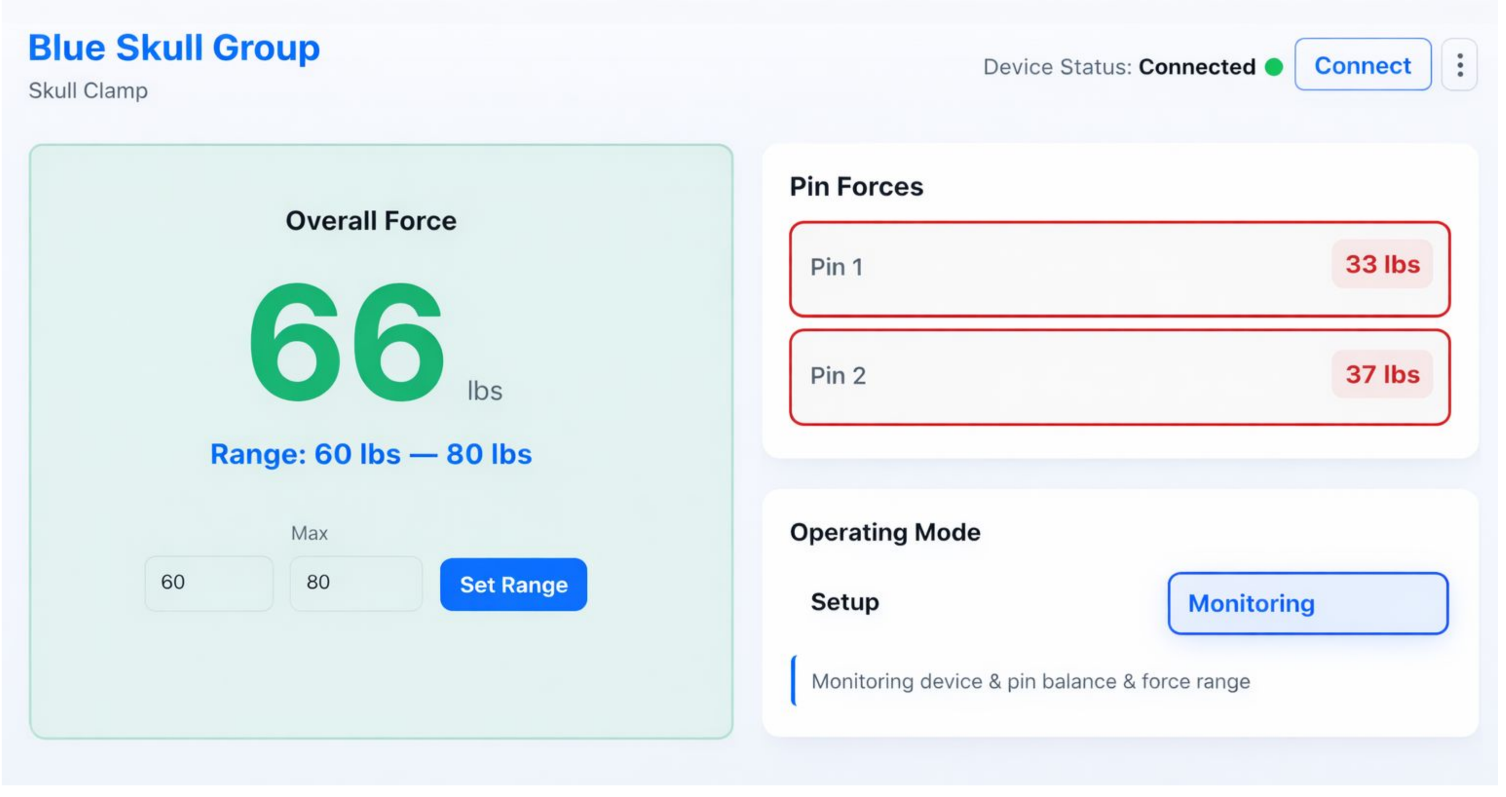

Figure 5. UI example: total force within range, but uneven pin forces.

Results

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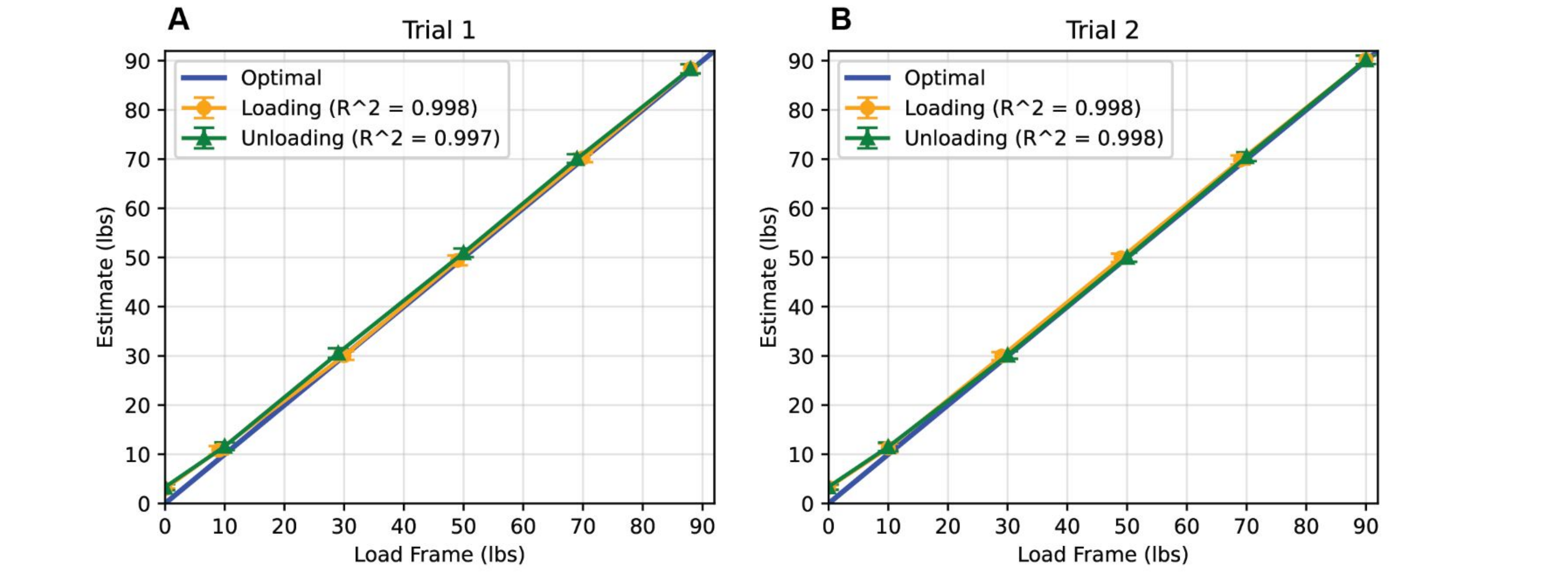

Figure 6. Accuracy of measurements remained consistent for forces between 0-90 lb. Measurements were linear and consistent across trials. No statistically significant effects due to hysteresis were seen. Bars indicate standard deviation of 120 samples.

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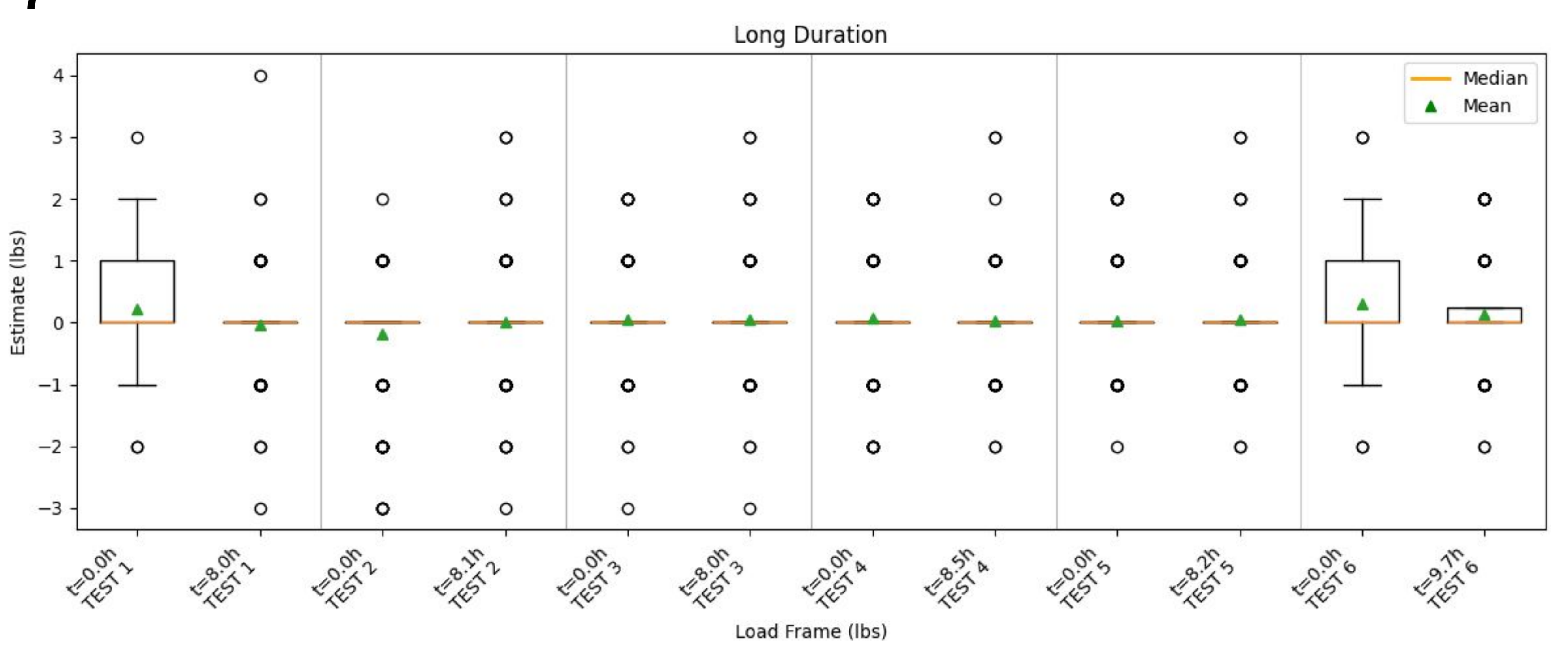

Figure 7. After eight hours of uninterrupted use, force measurements exhibited no statistically significant drift. The box plot summarizes the median and interquartile range (IQR), with whiskers extending to 1.5x IQR and individual markers indicated outliers among the 120 recorded samples.

Discussion

Our digital system enables the ability to **efficiently** and **accurately inform** neurosurgeons of the **force** exerted **on the skull** by the clamp.

Our device presents an improvement over existing analog indicators

- Easy to read, immediately accessible warnings of altered loading potentially indicative of situations requiring surgeon attention
- Forces resolved within ±1 lbf, with <2% error
- Measurement accuracy maintained following long durations of use

Stable per-pin monitoring is relevant for procedures where external perturbations can alter HC loading over time. Detecting real-time changes in force loading is an important capability not available in current devices.

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References

[1] Thijs, D., and Menovsky, T., 2021, “The Mayfield Skull Clamp: A Literature Review of Its Complications and Technical Nuances for Application,” World Neurosurgery, 151, pp. 102-109. https://doi.org/10.1016/j.wneu.2021.04.081.
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